Teacher:	Ray Joshua Balingkit	**Subject:**	Computer Science
Week:	2 (Term 1)	**Year:**	10-11
Link:		**Cover:**	

LESSON OVERVIEW

What should students know/do by the end of this week's lessons?

Students will develop their ability to represent and follow algorithms using flowcharts, pseudocode and trace tables. They will identify the purpose of common flowchart symbols and represent a decision with clearly labelled branches. Students will then follow instructions in order, record changing variable values and use trace evidence to explain a result or locate an incorrect prediction.

End-of-Week Assessment/Check: How will you know they got it?

Teacher questioning, discussion of worked examples, the completed student tasks and the final checks in both lessons. Check both branches and explain how the two representations match. Check every value change and require an explanation of the final output. Check that students can explain their reasoning and apply the skill without relying on the worked example.

LESSON SLOTS - MONDAY & FRIDAY

Lesson	Date	Learning Objective	The Lesson Flow	Differentiation
1	31 August 2026	By the end of this lesson, students will be able to represent a simple decision using a flowchart and pseudocode.	**H:** Compare a written recipe with a diagram of the same steps. Invite students to explain their prediction before demonstrating the example. Use their responses to identify the idea that needs clarification. **ID:** Introduce start/end, input/output, process and decision symbols; use labelled branches. Explain that a flowchart shows control flow while pseudocode expresses steps without requiring a particular programming language. A decision needs a precise condition and an outcome for each possible answer. **WD:** Convert an age-based ticket decision into both representations. Use a pass mark of 50 and trace scores of 49, 50 and 51. Ask students to choose the correct symbol for each step and label the	**Support:** Provide the flowchart symbols with their meanings and a partially completed decision. Use arrows and a condition word bank. Read the pseudocode aloud while students point to the matching flowchart step. **Extension:** Students add a permitted score range and explain how an invalid value should be handled before the pass decision. They justify why the order of the decisions matters.

			true and false routes before writing equivalent pseudocode. **YD:** Draw a flowchart and write matching pseudocode for deciding whether a score reaches a pass threshold. Students include input, a comparison, two clearly labelled outputs and an end point. Pairs trace the same three scores through both representations and correct any mismatch between the diagram and the written steps. **C:** Check both branches and explain how the two representations match. Ask students to support their answer with a specific example from their work, and record any misconception that needs a further demonstration.	
2	4 September 2026	By the end of this lesson, students will be able to trace an algorithm and record changes in variable values.	**H:** Predict the output of a short number-swapping algorithm. Invite students to explain their prediction before demonstrating the example. Use their responses to identify the idea that needs clarification. **ID:** Explain trace-table columns and updating values one instruction at a time. Explain that a trace table records execution rather than guesses about the intended result. Only update a value when an instruction changes it; the order of assignment statements therefore matters. **WD:** Trace a total and a discount decision using two supplied inputs. Use a price of 40 and quantity of 3, then apply a discount when the total exceeds 100. Record the total before and after the decision and compare it	**Support:** Provide named columns, line numbers and the first completed row. Highlight one instruction at a time and allow students to use value cards before transferring the changes into their trace table. **Extension:** Students create an input that follows the other branch and explain why its output differs. They identify a redundant table entry and distinguish unchanged values from uninitialised values.

			with a case that receives no discount. **YD:** Complete trace tables for two short algorithms, then identify and correct one incorrect predicted output. Students show the initial inputs, each changed value, the branch followed and the final output. They use the completed trace to explain the faulty prediction and rewrite it with a reason. **C:** Check every value change and require an explanation of the final output. Ask students to support their answer with a specific example from their work, and record any misconception that needs a further demonstration.	
3				Support: Extension:
4				Support: Extension:

MATERIALS NEEDED FOR THE WEEK

- Pearson Edexcel International GCSE Computer Science Student Book: Chapters 1 and 2
- Teacher computer and classroom display
- Classroom computers with Python 3 editor and individual save folders
- Prepared algorithm, tracing or Python practice tasks aligned with the lesson objectives
- Whiteboard and markers
- Student notebooks and pencils
- Success criteria, test tables and teacher observation checklist



END-OF-WEEK REFLECTION

- What worked well?

- What flopped or confused students?

- What needs to be re-taught or adjusted next week?